GG-S-150a
September 21, 1964
SUPERSEDING
Int. Fed. Spec. GG-S-00150(GSA-FSS)
July 30, 1963

FEDERAL SPECIFICATION

SCALE, AREA (TRANSPARENT PLASTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers transparent plastic scales for measuring and estimating areas of land in feet, rods, acres or miles and fractions thereof, and for use in map estimating of land in agriculture and soil conservation.

1.2 **Classification.**

1.2.1 *Types.* The sales shall be of the following types as specified in the invitation for bids (see 6.1).

Type I—Land area measurement by ratio of circle and grid method, in addition to linear ratio scales (figure 1).

Type II—Land area measurement by ratio of dotted square method, in addition to linear ratio scales (figures 2 and 3).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Clear, transparent, acrylic or vinyl plastic shall be used in manufacture of the scales.

3.2 **Construction.** Scales shall be thin, flat (in cross-section) and have a rectangular shape with rounded corners.

3.3 **Dimensions.**

Type I—

Overall length—approximately 5¼ inches.

Overall width—approximately 2⅜ inches.

Thickness—approximately 1/32 inch.

Rounded corners—approximate radius ⅛ inch.

Type II—

Overall length approximately 6⅜ inches.

Overall width approximately 2½ inches.

Overall thickness approximately 1/32 inch.

Rounded corners approximate radius ⅛ inch.

3.4 **Graduations and markings.** All graduations and markings shall be permanent, accurate and colored to be clearly legible and visible against background.

3.4.1 *Type I.* Graduations and markings on type I area scales shall include the following scales and ratios (see figure 1).

Linear scales:

10 scale (5 inches long) divided in 1/10 of an inch and numbered every inch.

1320 scale—1 inch = 1320 feet, divided in 100 feet and numbered every 400 feet.

*660 scale—1 inch = 660 feet, divided in 50 feet and numbered every 200 feet.

*Mile scale—8 inches = 1 mile, divided and numbered at ⅛, ¼ and ½ miles.

Area scales:

A circle to indicate each of the areas of .1, .2, .4, .6, .8, 1.0, 1.2, 1.5, 2.0, 2.5 and 3.0 acres. For direct acreage readings on a 600 scale (figure 1).

Multiple squares to indicate .1, .5, 1.0, 2.5, 5 and 10 acres. Square measure ratio of 1 square inch = 10 acres (figure 1).

3.4.2 *Type II.* Graduations and markings on type II area scales shall include the following scales and ratios (see figures 2 and 3).

Linear scales:

10 scale (6 inches long) divided in 1/10 of an inch and numbered every inch (figure 2).

*The 660 scale and mile scale may be combined into one.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Rod scale—1 inch = 40 rods, divided every 1 rod and numbered every 10 rods (figure 3).

44 scale—1 inch = 440 feet = 1/12 mile, divided every 10 feet and numbered every 100 feet (figure 2).

66 scale—1 inch = 660 feet = 1/8 mile, divided every 10 feet and numbered every 100 feet (figure 3).

Linear scales shall have white or near white non-transparent backgrounds for good visibility of graduations and numbers.

Area scales:

Eight 1 inch squares. Each square containing 100 dots represents 10 acres. Each dot represents 1/10 acre.

(Based on 660 scale—see figures 2 and 3).

Sixteen 1/2 inch squares. Each square containing 25 dots represents 2.5 acres. Each dot represents 1/10 acre. (Based on 660 scale—See figures 2 and 3).

The end margins on type II scales shall have white or near white non-transparent backgrounds, printed with "SCS".

"At 440 each dot equals .044 acre"

"At 660 each dot equals 1/10 acre"

(See figures 2 and 3).

3.5 Workmanship. The scales shall be free from defects which affect appearance or which might impair serviceability. The scales shall have no rough edges or burrs and the working edges shall be straight.

3.6 Identification marking. The manufacturer's name, trade-mark or other identifying information shall be marked on the scales.

3.7 Pocket clip. A metal pocket clip shall be provided on type I scale, located near one end as indicated in figure 1.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Samples for inspection shall be in accordance with MIL–STD–105.

4.3 Inspection. Samples selected in accordance with 4.2, shall be inspected for the following defects. Lot acceptance or rejection shall be based on acceptable quality level (AQL) of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

Classification of defects:

Major

(a) Material not as specified.

(b) Scales not accurate, twisted or does not lie flat.

(c) Graduations or markings not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Minor:

(a) Identification not as specified.

(b) Dimensions not as specified.

(c) Pocket clip not as specified.

4.4 **Sampling.** Sampling for test shall be in accordance with MIL–STD–105.

4.4.1 *Test.* Samples selected in accordance with 4.4 shall be tested as specified in 4.4.2. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

4.4.2 *Test.* A clean white cloth shall be moistened with water and rubbed under pressure, along the full length of one of the graduated surfaces of the scale, keeping the same portion of the cloth in contact with the scale at all times. The graduations and markings shall be rubbed not less than ten times in each direction. Any adverse effect on the graduations or markings, or any traces of coloring matter on the cloth shall constitute failure of this test.

4.5 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table I. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Sampling shall be in accordance with MIL–STD–105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE I.—*Classification of preparation for delivery defects.*

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 **Packaging.** Packaging shall be level A or C as specified (see 6.1).

5.1.1 *Level A.* Each scale shall be placed in an end-opening glassine or kraft envelope. Six scales of one type and description shall be packaged in a box conforming to PPP–B–566, style II, type A with pye locks, class a, and the box securely closed to prevent accidental opening.

5.1.1.1 *Intermediate packaging.* Unless otherwise specified (see 6.1) twelve packages of scales (72 scales), of one type and description, shall be intermediate packaged in a box conforming to PPP–B–566, style II, type A with pye locks, class a, and the box securely closed to prevent accidental opening. When specified (see 6.1) the box shall be waterproofed in accordance with MIL–P–116, method 1C–2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

5.1.2 *Level C.* The scales shall be packaged in accordance with the suppliers standard commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 *Level A.* Scales of one type and description, shall be packed in boxes conforming to PPP–B–636, class 2, V3c, style RSC. The gross weight of the box shall not exceed the weight limitations of the box specification. Each box shall be closed, waterproof sealed with tape and reinforced with bands of tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* Eight intermediate packages of scales (576 scale) shall be packed in a box conforming to PPP–B–636, type I, class 1, style RSC. Sealing and closure shall be in accordance with the appendix to the box specification. When quantities less than specified are required (see 6.1) the scales shall be packed in a box conforming to that specified herein.

5.2.3 *Level C.* The scales shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 **Standard Pack (civil agency procurements).** The Standard pack for civil agency procurements shall be as specified in 5.1.1, 5.1.1.1, and 5.2.2 (see 6.4).

5.4 **Marking.**

5.4.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, symbol and date of this specification.

(b) Type I or type II (see 1.2.1).

(c) Level of packaging and level of packing required (see 5.1 and 5.2).

(d) When quantities less than specified are required (see 5.2.2).

(e) When waterproofing of intermediate package is required (see 5.1.1.1).

(f) When inspection records are required (see 4.1).

6.2 Figures 1, 2 and 3 show examples of scales which have been found acceptable. However, figures 1, 2 and 3 are included for illustration purposes only and are not intended to preclude the furnishing of other designs, which conform to this specification.

6.3 Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Dimension scales, calculating, other than log or lumber rules.

Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Dimension scales, calculating, other than log or lumber rules.

Truckload minimum weight 10,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 **Standard Pack for civil agencies.** The standard pack requirements in 5.3 are intended for use in procurement of Stores Stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue and shipment to eventual user.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Figure 1.

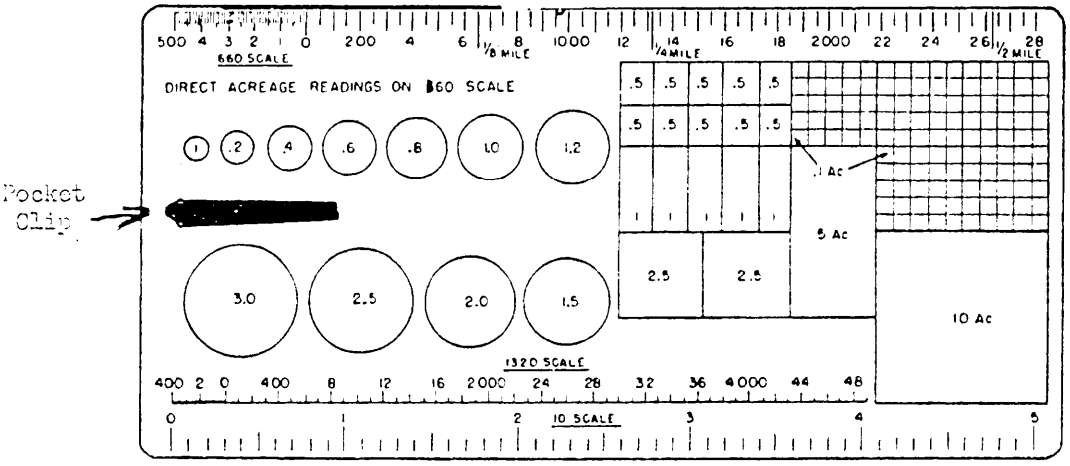


Figure 2.

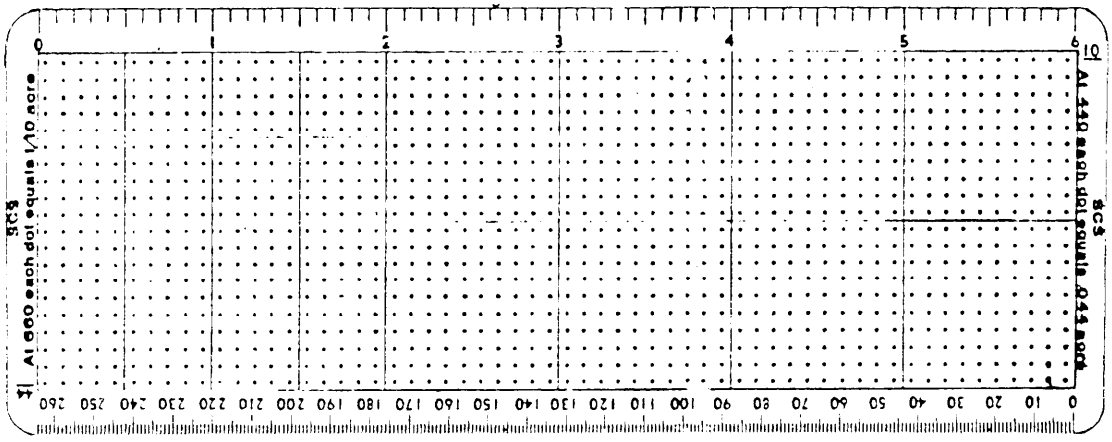
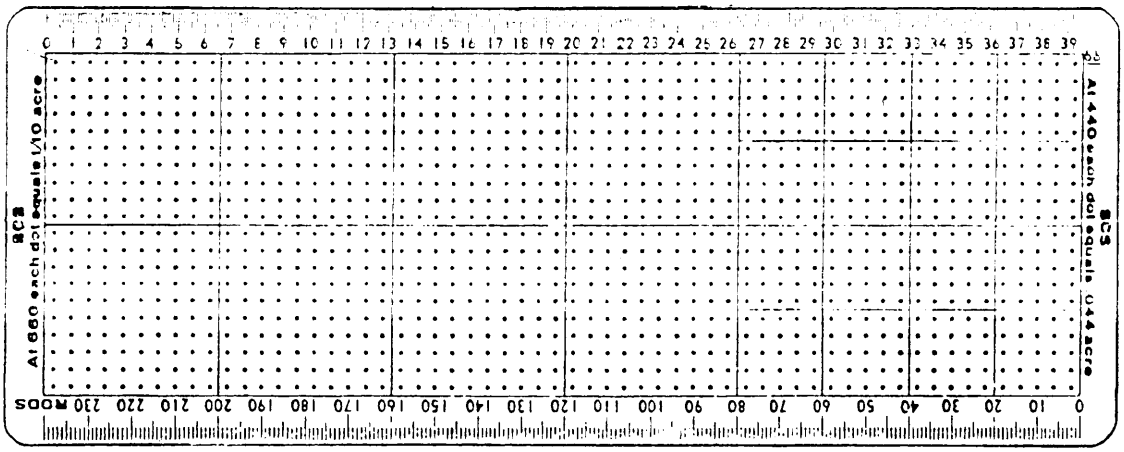


Figure 3.



Copies of this specification may be purchased for 5 cents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.